H-B-001137(GSA-FSS)
June 8, 1966

INTERIM FEDERAL SPECIFICATION

BRUSH, SCRUB, ALL-PLASTIC, STEAM KETTLE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an all-plastic steam kettle scrubbing brush with a separate handle of one type and grade.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H-S-951 -- Synthetic Filaments for Brushes; General Specification for.
PPP-B-636 -- Box, Fiberboard.
PPP-F-320 -- Fiberboard, Sheet and Stock and Cut Shapes.

Federal Standards:

Fed. Std. No. 123 -- Marking for Domestic Shipment.
Fed. Std. No. 406 -- Plastics: Methods of Testing: Federal Test Method Standard.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

FSC 7920

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing and Materials Standards

D256-56	--	Test for Impact Resistance of Plastics and Electrical Insulating Materials
D570-63	--	Test for Water Absorption of Plastics
D618-61	--	Conditioning Plastics and Electrical Insulating Materials for Testing
D638-61T	--	Test for Tensile Properties of Plastics (Tentative)
D648-56	--	Test for Deflection Temperature of Plastic Under Load
D790-63	--	Test for Flexural Properties of Plastics
D1525-58T	--	Test for Vicat Softening Point of Plastic (Tentative)

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

NOTE: Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS.

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample brush covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

3.2 Standard product. The brushes delivered under this specification shall be the manufacturer's standard commercial product except for any design modification and additional requirements specified herein.

3.3 Materials. Materials shall conform to the requirements specified hereinafter. Materials not definitely specified shall be of the quality used for the purpose in commercial practice.

3.3.1 Block. The block shall be made from a white opaque, high impact resistant, highly rubber-modified polystyrene thermoplastic with properties conforming to table 1 when tested as specified in 4.4.2.

Table I. Physical properties of plastic block material.

Property	Requirement
Tensile strength	1500 PSI (minimum)
Softening point (Vicat)	180°F (minimum)
Impact strength (Izod)	2.5 ft-lbs/in. of notch (minimum)
Water absorption	0.3% (Maximum)

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3.3.2 Filaments (bristles). Filaments shall be level, crimped, and natural color, made of a synthetic polymeric amide, conforming to class N of H-B-951. The diameter, crimp and crimp amplitude of the filaments shall conform to table II when measured as specified in 4.4.3.

Table II. Filament Dimensions.

Measure	Requirement
Diameter	0.022 inch plus or minus 0.0015 inch
Crimps per inch	6 (min) to 7-1/2 (max.)
Crimp amplitude	0.012 inch plus or minus 0.0015 inch

3.3.3 Handle. The handle shall be made from virgin white opaque, rigid, polyvinyl chloride compound with properties conforming to table III when tested as specified in 4.4.4.

Table III. Physical properties of polyvinyl chloride compound.

Property	Requirement
Tensile strength, psi, minimum	7,000
Modulus of elasticity in tension, psi, minimum	400,000
Flexural strength, psi, minimum	11,000
Izod impact strength, ft-lb/in. of notch, maximum	1.5
Deflection temperature @ 264 psi, deg. C., minimum	70

3.3.4 Handle grip. The handle grip shall be made from white opaque polyethylene or polyvinyl chloride compound normally used by the manufacturer for his commercial handle grip.

3.3.5 Staple wire. The staple wire shall be a commercial steel staple wire, plated, or coated to resist corrosion, or a commercial grade of corrosion-resisting steel wire. The diameter shall be not less than 0.039 inch.

3.4 Construction. The finished brush head shall be cylinderical shaped with spherical shaped ends similar to figure 1. The brush head, handle, and handle grip shall conform to the requirements specified herein and in table IV.

Table IV. Component requirements.

Component Part	Required
Block:	
Length, o/a (minimum inches)	4-13/16
Diameter (minimum inches)	2-1/4
Tuft holes	
Depth (plus or minus 1/16 inch)	5/16
Diameter (average)	15/64
Number (minimum)	
cylindrical body	112
each end	41
Handle hole	
Depth (minimum inches)	1-1/2
Diameter (approximate inches-large end)	13/16
Taper (approximate)	2°

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Table IV. Component requirements. (cont'd)

Component Part	Required
Filaments (bristles):	
Length clear of block (minimum inches)	1-5/16
Length brush face, o/a (minimum inches)	7-1/2
Width brush face, o/a (minimum inches)	5
Number (ends) per tuft hole	
cylindrical body (plus or minus 4)	56
each end (plus or minus 4)	50
Handle:	
Length, o/a (minimum inches)	36
Outside diameter (minimum inches)	31/32
Wall thickness (minimum inches)	7/32
Degree of taper to fit handle hole	------
Diameter of taper (small end-approximate inches)	11/16
Straightness (bow and/or warp maximum inches)	1/4
Handle grip:	
Length, o/a (minimum inches)	5-1/8
Diameter hanging hole (minimum inches)	5/16
Inside diameter must be snug fit on handle	------

3.4.1 Block. The block shall be smoothly finished one piece construction made from the thermo plastic specified in 3.3.1. The hole for the handle shall be located an equidistant from each end of the block and the tuft holes (table IV) shall be located in accordance with the manufactuere's standard commercial practice. An area approximately 2 by 4 inches located on the handle hole center lines shall be free of tuft holes. The 4 inch dimension shall be parallel to the length of the block.

3.4.2 Filaments (bristles). The filaments shall solidly fill the tuft holes in the block and shall be securely staple set with wire staples (3.3.5). The staples shall straddle over the middle of the tuft and shall be driven into the block at the bottom of the holes so the tuft shall not separate from the block when tested as specified in 4.4.1.

3.4.3 Handle. The handle shall be a thick walled tube manufactured from the polyvinyl chloride specified in 3.3.3. The dimensions shall be as specified in table IV. The handle shall be tested for straightness as specified in 4.4.5 and shall not split, crack, or break when flattened as specified in 4.4.6.

3.4.4 Handle grip. The handle grip shall be one piece construction manufactured from the polyethylene or polyvinyl specified in 3.3.4. It shall have a ribbed or similar type gripping surface and a hanging hole. It shall not turn or pull freely from the handle (3.4.3).

3.5 Finish. The brush blocks, handles, and handle grips shall be finished in accordance with the manufacturer's standard commercial practice.

3.6 Marking for identification. Unless otherwise specified (see 6.2) in addition to any special marking required by the contract or order, the manufacturer shall permanently and legibly indent his name, or trademark, of such known character easily indentifiable with said manufacturer, on each brush block.

3.7 Workmanship. The brush blocks, handles, and handle grips shall be free from defects which affect their appearance or which affect their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for the defects in table VI and specified dimensions. The preproduction sample shall be tested in accordance with 4.4.1, 4.4.5 and 4.4.6.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 In process inspection. During production, an in process inspection shall be made for defects that cannot be inspected after assembly. The inspection lot shall consist of all brush blocks being assembled for delivery to the Government. The lot size shall be expressed in units of brush block ready for the filament to be assembled. The samples shall be examined for the defects listed in table V. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.0 expressed in terms of defects per hundred units.

Table V. Defects in process inspection.

Examine	Defect
Length block	Less than 4-13/16 inch
Diameter block	Less than 2-1/4 inch
Depth handle hole	Less than 1-1/2 inch
Depth tuft holes	Less than 1/4 inch
Diameter tuft holes	Less than 15/64 inch average
Number of tuft holes:	
Cylindrical part	Less than 112
Each end	Less than 41
Staple wire	Less than specified gauge

4.3.3 End item inspection. The lot shall consist of all brushes and handles with grips offered for inspection at one time. The sample unit shall be one completely fabricated brush, handle, and handle grip.

4.3.3.1 Visual examination. Examination shall be made of the end item for the defects in table VI. The inspection level shall be level II with an AQL of 4.0 expressed in terms of defects per hundred units.

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Table VI. Visual defects.

Examine	Defect
Finish	Finish not representative of a quality commercial product.
Construction and workmanship	Rough or cracked surfaces. Handles and handle grips have mold fins, nodes, etc. that will cause discomfort to hands. Filler filament dirty or imbedded with foreign matter. Filler holes not fully filled with filament. Handle does not fit block properly. Handle grip does not fit handle properly.
Markings for identification	Omitted, illegible. Not indented. Special markings not provided when specified.

4.3.3.2 Dimensional examination. Inspection shall be made of the end items for compliance with the dimensions specified in table IV. Any deviation from the specified requirements shall constitute a defect. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item. The end item shall be tested as specified in 4.4.1, 4.4.5, and 4.4.6. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

4.3.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VII. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

Table VII. Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior)	Omitted, incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, or distortion of container.
Contents (interior)	Number per container is more or less than required. Net weight exceeds requirements.

4.4 Tests.

4.4.1 Set retention. Grasp the individual tuft of filaments (bristles) with a suitable holding device and apply a pulling force, with a spring scale or a weight, of eight pounds in a direction tending to pull the tuft from the block to determine compliance with 3.4.2. Ten determinations shall be made per brush. Failure of one or more determinations shall constitute a defect.

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4.4.2 Block material. The thermoplastic material used to fabricate the blocks shall be tested in accordance with Federal Standard No. 406 test methods or the ASTM test methods indicated in table VIII. A sample of the thermoplastic shall be selected and compression molded into the required test specimens. The number and size of the prepared specimens shall be suitable for conducting the test specified. The results obtained from five (5) tests for each characteristic shall be averaged to determine compliance with 3.3.1. Failure of any characteristic shall be cause for rejection of the lot of thermoplastic material.

Table VIII. Test methods for thermoplastic.

Characteristics	Fed. Std. 406 Test Method No.	ASTM Test Method
Softening point (vicat)	----	D1521-58T
Impact strength (Izod)	1071	D256 -56
Tensile strength	1011	D638 -61T
Water absorption	7031	D570 -63

4.4.3 Filaments. To determine compliance with 3.3.2, the filament diameter, crimp (see 4.4.3.1) and crimp amplitude shall be measured as described in H-S-951.

4.4.3.1 Crimp alternate method. When the apparatus specified in H-S-951 is not available to measure the crimp the following method shall be used: Ten (10) filaments selected at random shall be measured. A ruler graduated in 32nds of an inch shall be used to measure the crimps. Each filament shall be positioned with one side of the crimp touching the ruler and the trough of the crimp at the inch mark. Count the number of troughs touching the ruler for a distance of one inch. The readings shall be averaged. The average value shall meet the requirement in 3.3.2.

4.4.4 Handle material. The polyvinyl chloride compound used to fabricate the handles shall be tested in accordance with Federal Standard No. 406 test methods or the ASTM test methods specified in 4.4.4.1 through 4.4.4.6. A sample of the polyvinyl chloride shall be selected and compression molded into the required test specimens. The number and size of the prepared specimens shall be as specified in each test procedure. The results obtained from the tests for each characteristic shall be averaged to determine compliance with 3.3.3. Failure of any characteristic shall be cause for rejection of the lot of polyvinyl chloride.

4.4.4.1 Conditioning. The test specimens for deflection temperature under load shall be conditioned in accordance with the general requirements of Fed. Test Std. No. 406 or Procedure B of ASTM Method D618, except that the minimum conditioning time shall be 24 hours. All other molded test specimens shall be conditioned in accordance with Method A of ASTM Method D618, except that the minimum conditioning time shall be 24 hours.

4.4.4.2 Test conditions. Unless otherwise specified, all tests shall be conducted at 23 $\pm$ 1°C. (73.4° $\pm$ 1.8°F.) and 50 $\pm$ 2 percent relative humidity. The average values shall be reported.

4.4.4.3 Tensile strength and modulus of elasticity in tension. Five specimens 1/8 inch thick, shall be tested in accordance with method 1011 of Fed. Test Std. No. 406 or ASTM Method D638-61T.

4.4.4.4 Flexural strength. A minimum of five specimens 1/8 by 1/2 by 4 inches shall be tested in accordance with method 1031 of Fed. Test Std. No. 406, or method B of ASTM D790-63, using a span of two inches and a cross head speed of 0.50 inch per minute.

4.4.4.5 Izod impact strength. Five specimens, each with a thickness between 1/4 inch and 1/2 inch shall be tested in accordance with method 1071 of Fed. Test Std. No. 406 or method A of ASTM D256-56. If it is found that specimens in the lower part of the 1/4 to 1/2 inch thickness range produce incomplete breaks while those in the upper part of this range show complete breaks, the values obtained with the thicker specimens shall be reported.

4.4.4.6 Deflection temperature under load. A minimum of two specimens 1/4 inch thick shall be conditioned in accordance with method 2011 of Fed. Test Std. No. 406 or method B of ASTM Method D648-56 except that the minimum conditioning time shall be 24 hours; the two specimens shall then be tested at a fiber stress of 264 psi in accordance with ASTM Method D648-56.

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4.4.5 Straightness (finished handle). Test for straightness by placing the handles on a flat, smooth surface in a horizontal plane. Roll each handle 360°, using light finger pressure, to determine the maximum departure of the handle from the flat surface. Measure the distance of departure. (See 3.4.3 and table IV).

4.4.6 Flattening (finished handle). Reduce a 2 inch long (plus 1/4 inch) handle specimen's outside diameter 25 to 30 percent (e.g., 1 inch OD to 3/4 inch) between parallel plates of a press or vise. Examine the specimen for compliance with 3.4.3. Note: The test shall be completed within a maximum time of 5 minutes and the test specimens shall be excess stock left from the handles.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified.

5.1.1 Level A. Each brush shall be packaged in a plastic bag and sealed.

5.1.2 Level B. The brushes shall be packaged as specified in 5.1.1.

5.1.3 Level C. The brushes, handles and handle grips shall be packaged to afford adequate protection against deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. The suppliers commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve unit packages of brushes, handles and handle grips shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. Fiberboard cushioning and blocking shall be provided with the fiberboard meeting the requirements of PPP-F-320, class domestic, variety SW. The box shall be closed, waterproof sealed with tape in accordance with the appendix to the box specification.

5.2.2 Level B. Twelve unit packages of brushes, handles and handle grips shall be packed in a close fitting box conforming to PPP-B-636, class domestic. Fiberboard cushioning and blocking shall be provided with the fiberboard meeting the requirements of PPP-F-320, class domestic, variety SW. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The brushes, handles and handle grips shall be packed to insure carrier acceptance and safe delivery at destination at lowest rates in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brushes covered by this specification are intended for scrubbing large steam kettles and vats used in hospitals and similar institutions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a preproduction sample is required (see 3.1).
(c) Special markings required (see 3.6).
(d) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

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6.3 Illustration. The illustration shown herein as figure 1 is for the purpose of illustrating the general design and is not intended to preclude any brush which otherwise meets the requirements of this specification.

6.4 Weight of brush components. The average weight of the finished brush components are as follows:

Brush head	14 ounces
Handle and handle grip	17 ounces
Total weight	1 lb. 15 ounces

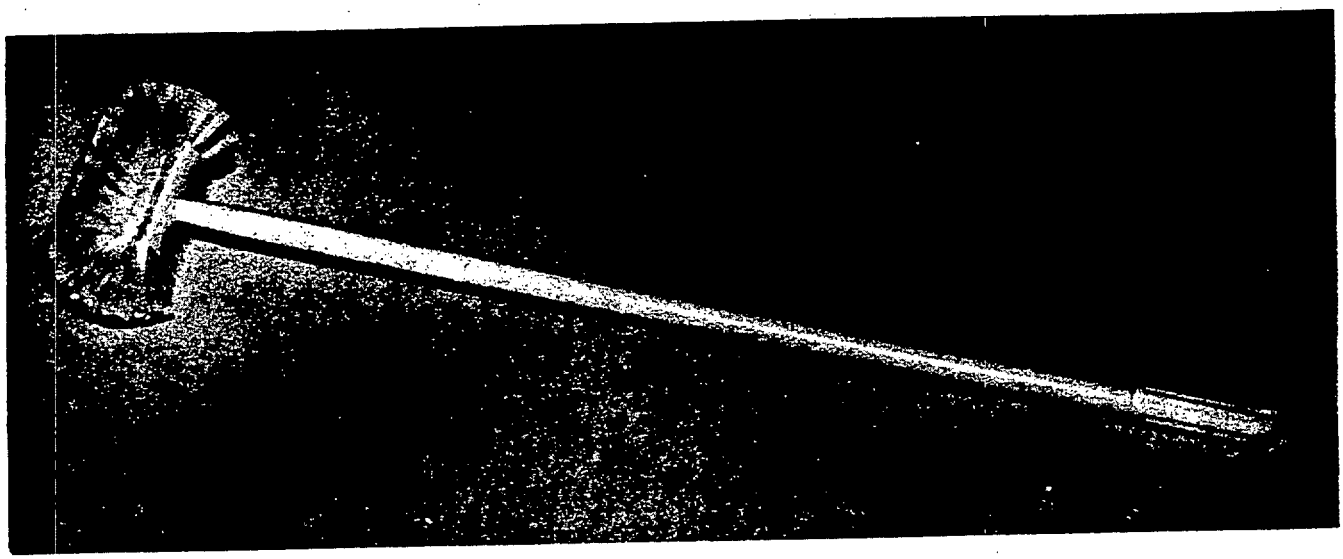


Figure 1

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